



DATA ANALYTICS PROJECT

Customer Shopping Behavior Analysis

End-to-End Data Pipeline: Python ETL → MySQL → SQL Analysis → Power BI Dashboard

DATA ANALYTICS

PYTHON · SQL · POWER BI

Project Overview

This project builds a complete, end-to-end data analytics pipeline for customer shopping behavior data. It covers the full lifecycle of a real-world workflow: extracting and cleaning raw transactional data with Python, loading it into a MySQL database, performing exploratory data analysis (EDA) and business-transaction queries directly in SQL, and finally connecting Power BI to the database to build a live, interactive dashboard.



3,900

Records

Customer purchases



18

Columns

Demographics & behavior

WHAT THIS COVERS



Extract & clean raw data



Centralize in MySQL



Query with SQL



Visualize in Power BI

A reproducible workflow from messy CSV to actionable business intelligence.

Tech Stack



Extraction & Transformation

Python (pandas)

Load raw CSV, clean, impute, engineer features



Database

MySQL

Central store for cleaned data; source of truth



Database Connectivity

**mysql-connector-python /
SQLAlchemy**

Connect Python to MySQL for loading data



Analysis

SQL (MySQL)

EDA and business-transaction queries



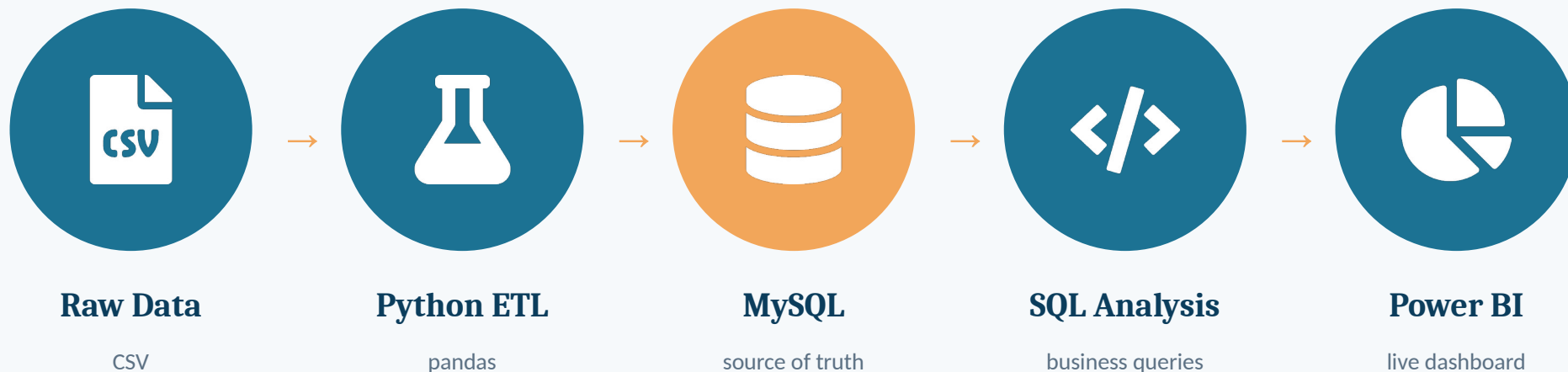
Visualization

Power BI Desktop

Interactive dashboard, connected via native SQL connector

Pipeline Architecture

A five-stage linear flow with MySQL at the center as the single source of truth for both SQL analysis and Power BI.



Each stage feeds the next, ensuring data quality and consistency from raw CSV through to a live, interactive dashboard — with no manual re-exports required.

Phase 1 — ETL Using Python



Data Loading

Imported dataset with pandas; used `df.info()` and `.describe()` for initial exploration.



Missing Data

Imputed 37 missing Review Rating values using the median rating per product category.



Standardization

Renamed columns to snake_case; dropped the redundant `promo_code_used` column.



Feature Engineering

Created `age_group` by binning ages; derived `purchase_frequency_days` from purchase data.

Phase 2 — Loading into MySQL

The cleaned DataFrame was loaded into MySQL so all subsequent analysis runs as SQL against a real database — not in-memory pandas. This also lets Power BI connect to the same source of truth.

1



Connect

Connected to MySQL from Python using SQLAlchemy / mysql-connector-python.

2



Create Schema

Built a table matching the cleaned schema (18 columns including age_group).

3



Load Data

Used `df.to_sql()` to load the cleaned DataFrame — no need to re-run cleaning for every analysis.

Phase 3 — SQL Analysis: Key Findings

Ten business-transaction queries run directly in MySQL.

\$157K

Male Revenue

vs. \$75K female — more than double

3,116

Loyal Customers

Vast majority of the base

~49%

Discount Dependency

Hat, Sneakers, Coat — margin risk

26.88%

Subscribers

Yet avg spend is flat vs. non-subscribers



Revenue is evenly spread across age groups — age is not a strong differentiator on its own.

Phase 4 — Power BI Dashboard



Power BI connected directly to MySQL via its native SQL connector — no static CSV imports. The dashboard stays synced with the same data used for SQL analysis.

Live • Import mode • auto-refresh from MySQL



KPI Cards

4K customers, \$233K revenue, 3.75 avg rating, \$60 avg purchase



Slicers

Subscription Status, Gender, Category, Shipping Type



DAX Measures

Built for all key metrics



Charts

Monthly revenue trend, category performance, payment methods, top cities, item colors, clothing sizes

Key Insights & Business Recommendations



Review Discount Policy

Hat, Sneakers, Coat are ~49–50% discount-dependent — test demand elasticity before cutting.



Focus on Retention

3,116 Loyal customers — invest in keeping them rather than early-funnel acquisition.



Targeted Marketing

Prioritize male customers and express-shipping users; they show the strongest revenue signals.



Re-examine Subscriptions

Avg spend is flat between subscribers and non-subscribers — the program isn't proven to drive higher spend.

Conclusion

This project demonstrates a complete analytics pipeline from raw data to a live business dashboard. Because Power BI reads directly from MySQL rather than a static file, any future data update automatically flows through to the dashboard — no manual re-export required.



Python ETL

Cleanse and validate raw data



MySQL Source

Central queryable source of truth



SQL Analysis

Structured business queries and metrics



Power BI Live

Real-time dashboards connected to MySQL



A reproducible, end-to-end workflow that scales — from messy CSV to actionable business intelligence.